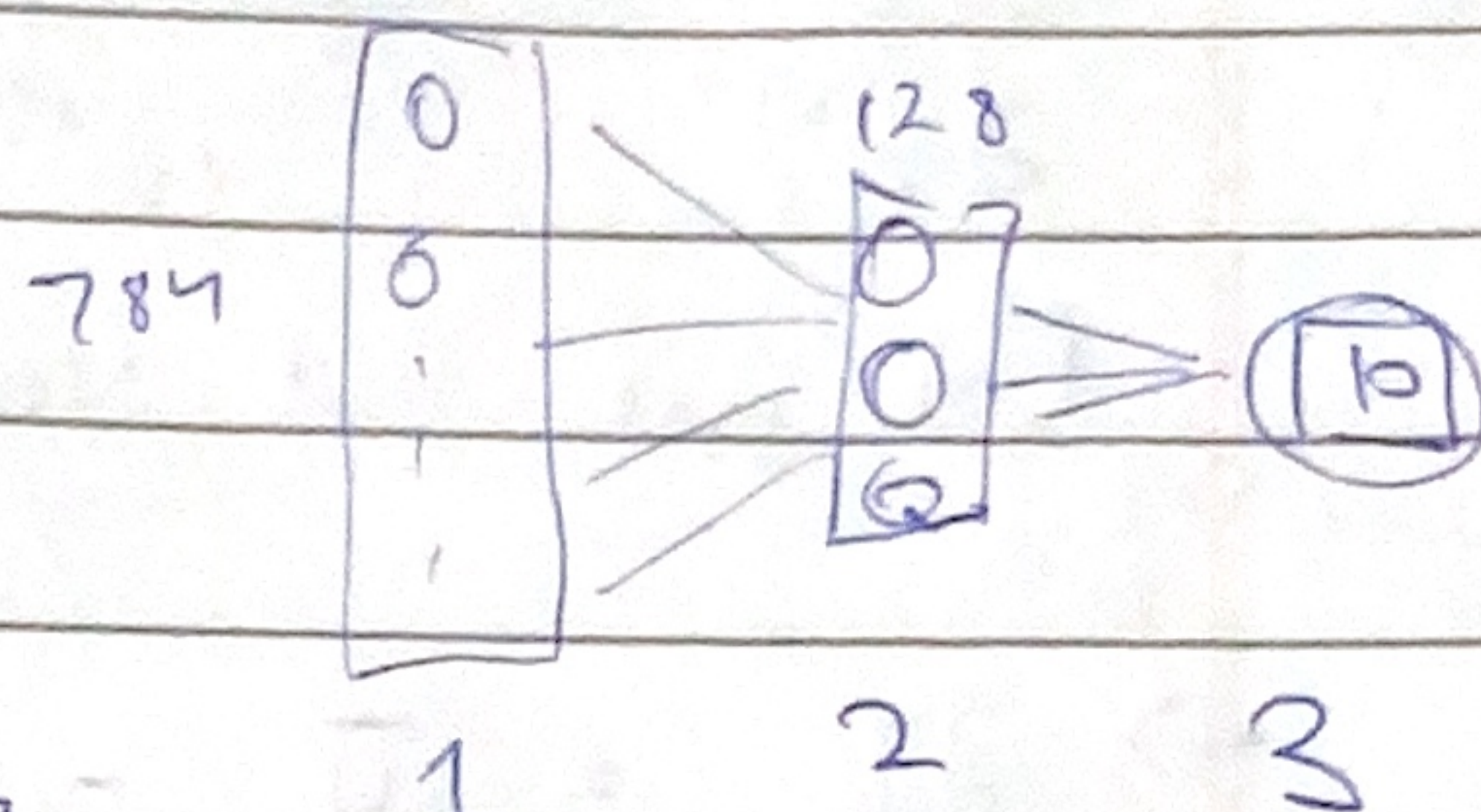


$$L = -\log \quad \Delta \phi_i = -L \frac{\partial L}{\partial \phi_i}$$

Binary Classifier

$$X \rightarrow \begin{bmatrix} 1 \\ h; 784 \end{bmatrix}$$



~~A^[1]~~

$$Z^{[2]} = W^{[2]} X^{[1]} + b^{[2]}$$

$$A^{[2]} = \text{ReLU}(Z^{[2]})$$

$$Z^{[3]} = W^{[3]} A^{[2]} + b^{[3]}$$

$$\bar{Y}_{\text{pred}} = \text{softmax}(Z^{[3]}) \quad \text{correct value}$$

$$X^{[1]} = A^{[1]}$$

inputs

$$\text{softmax} = \frac{e^{z_i}}{\sum e^{z_i}}$$

$$\text{Loss} = - \sum y_i \log(\bar{Y}_i)$$

$\bar{Y}$ is predicted, Y is correct value.

for one hot vectors, $\text{Loss} = -y_k \log(\bar{y}_k)$ where k is correct class

since in one hot, y_i where $i \neq k$ is zero.

$$L = - \sum y_i \log(\bar{y}_i) = - \sum y_i \log\left(\frac{e^{z_i}}{\sum e^{z_j}}\right)$$

$$\bar{y}_i = \frac{e^{z_i}}{\sum e^{z_j}} = \sum y_i (\log(\sum e^{z_j}) - \log(e^{z_i}))$$

$$L = \sum y_i (\log(\sum e^{z_j}) - z_i)$$

$$\frac{\partial L}{\partial z_i^{[3]}} = \sum y_i \left(\frac{\partial}{\partial z_i^{[3]}} \log(\sum e^{z_j}) - 1 \right) + y_i$$

$$L = \sum y_i (\log(\sum e^{z_j}) - z_i) \quad \text{For } i=1,$$

$$\frac{\partial L}{\partial z_1^{[3]}} = \left(y_1 \frac{e^{z_1}}{\sum e^{z_j}} + y_2 \frac{e^{z_1}}{\sum e^{z_j}} + \dots \right) - (y_1)$$

$$\Rightarrow \frac{\partial L}{\partial z_i^{[3]}} = \text{softmax}(z_i) \left(\sum y_j \right) - y_i$$

$$\frac{\partial L}{\partial z_i^{[3]}} = \text{softmax}(z_i^{[3]}) \left(\sum_{j=1}^n y_j \right) - y_i$$

For one-hot vectors, $\sum_{j=1}^n y_j = 1$ [0, 0, 1, 0, ...] ^{correct y_i}

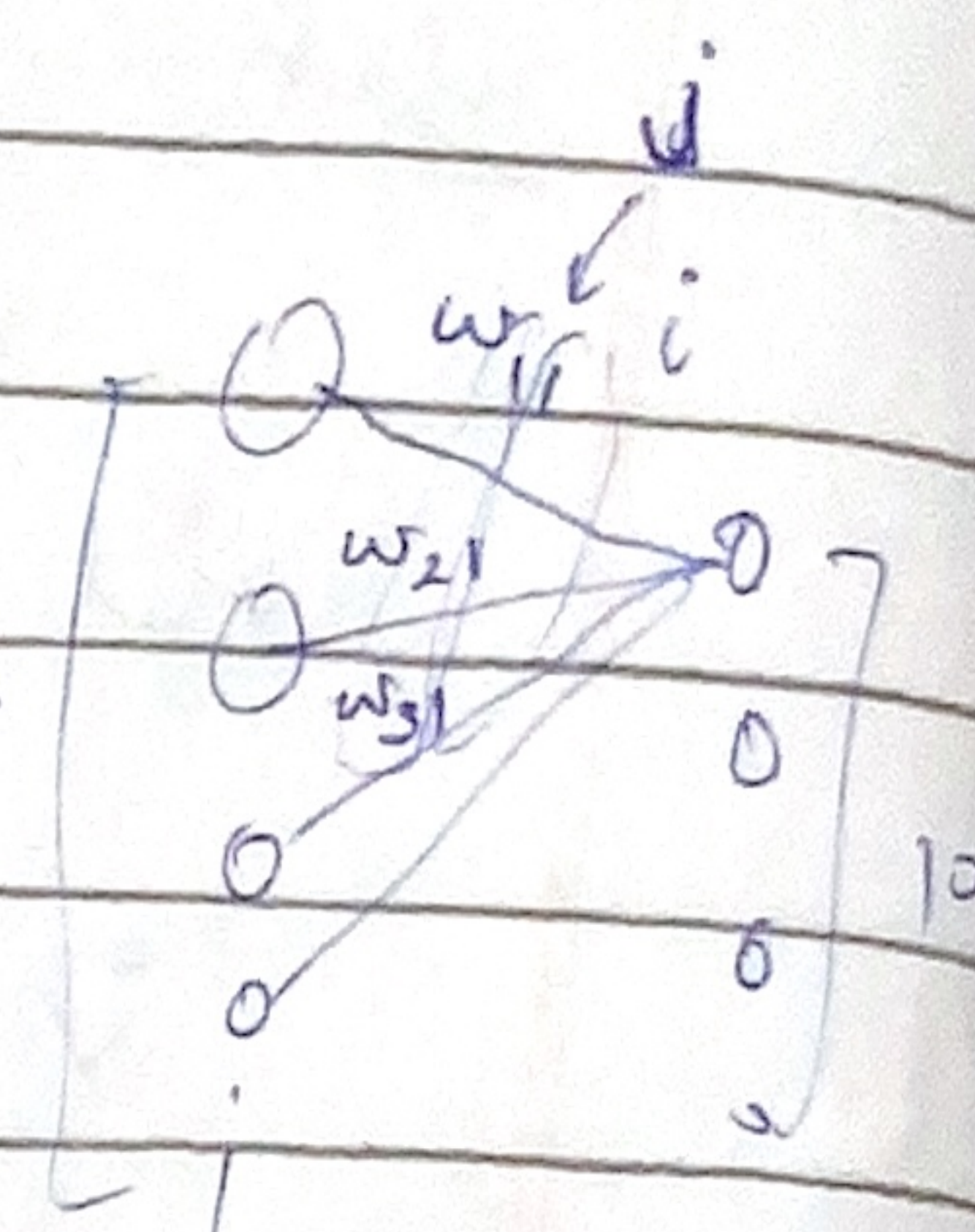
$$\Rightarrow \frac{\partial L}{\partial z_i^{[3]}} = \bar{y}_i - y_i \quad (\because \bar{y}_i = \text{softmax}(z_i^{[3]})) \quad [\text{prediction}]$$

$$= \begin{cases} \bar{y}_i & \text{when } i \neq k \\ \bar{y}_i - 1 & \text{when } i = k \end{cases} \quad \text{b/c } y_i = \begin{cases} 1 & i = k \\ 0 & i \neq k \end{cases}$$

Since $z^{[3]}$ is a $[10 \times 1]$ vector

$$\frac{\partial L}{\partial z^{[3]}} = \left[\frac{\partial L}{\partial z_1}, \frac{\partial L}{\partial z_2}, \dots, \frac{\partial L}{\partial z_{10}} \right] \quad (28)$$

$$\text{Now } z_i^{[3]} = \left(\sum_{j=1}^{128} w_{ji}^{[3]} a_j^{[2]} \right) + b_i^{[3]}$$



$$\frac{\partial L}{\partial b_i^{[3]}} = \frac{\partial L}{\partial z_i^{[3]}} \quad \frac{\partial L}{\partial w_{ji}^{[3]}} = \frac{\partial L}{\partial z_i^{[3]}} \times \frac{\partial z_i^{[3]}}{\partial w_{ji}^{[3]}} = \frac{\partial L}{\partial z_i^{[3]}} \times a_j^{[2]}$$

$$W^{[3]} = \begin{bmatrix} w_{11}^{[3]} & w_{12}^{[3]} & \dots & w_{1,10}^{[3]} \\ w_{21}^{[3]} & & & \\ \vdots & & & \\ w_{128,1}^{[3]} & & & w_{128,10}^{[3]} \end{bmatrix}$$

$$\frac{\partial L}{\partial W^{[3]}} = \begin{bmatrix} \frac{\partial L}{\partial w_{1,1}^{[3]}} & \dots & \frac{\partial L}{\partial w_{1,10}^{[3]}} \\ \vdots & & \vdots \\ \frac{\partial L}{\partial w_{128,1}^{[3]}} & \dots & \frac{\partial L}{\partial w_{128,10}^{[3]}} \end{bmatrix} = \begin{bmatrix} a_1^{[2]} & a_1^{[2]} & \dots & a_1^{[2]} \\ a_2^{[2]} & a_2^{[2]} & \dots & \\ \vdots & \vdots & & \\ a_{128}^{[2]} & a_{128}^{[2]} & \dots & a_{128}^{[2]} \end{bmatrix}$$

$$\langle \psi | \psi \rangle$$

$$| \text{state} \rangle$$

ket

$$P = |\psi|^2$$

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$$\frac{\partial L}{\partial w^{[3]}} = \begin{bmatrix} a_1^{[2]} \frac{\partial L}{\partial z_1^{[3]}} & a_1^{[2]} \frac{\partial L}{\partial z_2^{[3]}} & \dots & a_1^{[2]} \frac{\partial L}{\partial z_{10}^{[3]}} \\ a_2^{[2]} \frac{\partial L}{\partial z_1^{[3]}} & a_2^{[2]} \frac{\partial L}{\partial z_2^{[3]}} & & \\ \vdots & \vdots & & \\ a_{128}^{[2]} \frac{\partial L}{\partial z_1^{[3]}} & a_{128}^{[2]} \frac{\partial L}{\partial z_2^{[3]}} & & a_{128}^{[2]} \frac{\partial L}{\partial z_{10}^{[3]}} \end{bmatrix}$$

Now $a_i^{[2]} = \text{ReLU}(z_i^{[2]})$ $z_i^{[2]} = \left(\sum_{j=1}^{128} w_{ji}^{[2]} a_j^{[1]} \right) + b_i^{[2]}$

$$a_i^{[2]} = \begin{cases} z_i^{[2]} & \text{when } z_i^{[2]} \geq 0 \\ 0 & \text{when } z_i^{[2]} < 0 \end{cases}$$

$$\frac{\partial a_i^{[2]}}{\partial z_i^{[2]}} = \begin{cases} 1 & \text{when } z_i^{[2]} > 0 \\ 0 & \text{when } z_i^{[2]} < 0 \end{cases} \quad \text{Not differentiable at } \underline{0}$$

$$\frac{\partial L}{\partial w^{[2]}} = \begin{bmatrix} a_1^{[1]} & \dots & a_1^{[1]} \\ a_2^{[1]} & & \\ \vdots & & \\ a_{784}^{[1]} & \dots & a_{784}^{[1]} \end{bmatrix}$$

$$\frac{\partial L}{\partial w^{[2]}} \neq \frac{\partial L}{\partial w^{[2]}}$$

$$A^{[2]} \rightarrow (1 \times 128)$$

$$\frac{\partial L}{\partial w^{[3]}} = \begin{bmatrix} a_1^{[2]} \\ a_2^{[2]} \\ \vdots \\ a_{128}^{[2]} \end{bmatrix} \begin{bmatrix} \frac{\partial L}{\partial z_1^{[3]}} & \frac{\partial L}{\partial z_2^{[3]}} & \dots & \frac{\partial L}{\partial z_{10}^{[3]}} \end{bmatrix}$$

$$= (A^{[2]})^T \left(\frac{\partial L}{\partial z^{[3]}} \right)$$

$$\frac{\partial L}{\partial b^{[2]}} = \frac{\partial L}{\partial z^{[2]}} = \frac{\partial L}{\partial z^{[3]}} \times \frac{\partial z^{[3]}}{\partial a^{[2]}} \times \frac{\partial a^{[2]}}{\partial z^{[2]}}$$

$$= \begin{cases} \frac{\partial L}{\partial z^{[3]}} \times \frac{\partial L}{\partial w^{[3]}} & z_i^{[2]} \geq 0 \\ 0 & z_i^{[2]} < 0 \end{cases}$$

ambiguous

$$\frac{\partial L}{\partial z^{[3]}} \rightarrow 1 \times 10$$

(1 2 3 ...)

$$z_i^{[3]} = \sum_{j=1}^{128} w_{ji}^{[3]} a_j^{[2]} + b_i^{[3]}$$

$$a_j^{[2]} = \text{ReLU}(z_j^{[2]})$$

$$z_j^{[2]} = \sum_{k=1}^{784} w_{kj}^{[2]} a_k^{[1]} + b_j^{[2]}$$

$$\frac{\partial z_j^{[2]}}{\partial b_j^{[2]}} = 1 \quad \frac{\partial L}{\partial b_j^{[2]}} = \frac{\partial L}{\partial z_i^{[3]}} \times \frac{\partial z_i^{[3]}}{\partial a_j^{[2]}} \times \frac{\partial a_j^{[2]}}{\partial z_j^{[2]}} \times \frac{\partial z_j^{[2]}}{\partial b_j^{[2]}}$$

error error error error

~~$$\frac{\partial L}{\partial b_j^{[2]}} = \frac{\partial L}{\partial z_i^{[3]}}$$~~

$$\frac{\partial L}{\partial b_j^{[2]}} = \left(\sum_{i=1}^{10} \frac{\partial L}{\partial z_i^{[3]}} \frac{\partial z_i^{[3]}}{\partial a_j^{[2]}} \right) \times \frac{\partial a_j^{[2]}}{\partial z_j^{[2]}} \times 1$$

$$\frac{\partial L}{\partial b_j^{[2]}} = \left(\sum_{i=1}^{10} \frac{\partial L}{\partial z_i^{[3]}} w_{ji}^{[3]} \right) \frac{\partial a_j^{[2]}}{\partial z_j^{[2]}}$$

δ_j

$$\frac{\partial a_j^{[2]}}{\partial z_j^{[2]}} = \begin{cases} 1 & z_j^{[2]} > 0 \\ 0 & z_j^{[2]} < 0 \end{cases}$$

$$\frac{\partial L}{\partial b_1^{[2]}} = \left(\frac{\partial L}{\partial z_1^{[3]}} w_{11}^{[3]} + \frac{\partial L}{\partial z_2^{[3]}} w_{12}^{[3]} + \dots + \frac{\partial L}{\partial z_{10}^{[3]}} w_{1,10}^{[3]} \right) \frac{\partial a_1^{[2]}}{\partial z_1^{[2]}}$$

$$\frac{\partial L}{\partial b_2^{[2]}} = \left(\frac{\partial L}{\partial z_1^{[3]}} w_{2,1}^{[3]} + \dots + \frac{\partial L}{\partial z_{10}^{[3]}} w_{2,10}^{[3]} \right) \frac{\partial a_2^{[2]}}{\partial z_2^{[2]}}$$

$$\frac{\partial L}{\partial b^{[2]}} = \left[\frac{\partial L}{\partial z^{[3]}} \cdot \frac{\partial z^{[3]}}{\partial a^{[2]}} \right] \times \left[\frac{\partial a^{[2]}}{\partial z^{[2]}} \right]$$

(1 × 10) (10 × 784) element wise multiply

$$\frac{\partial L}{\partial b^{[2]}} = \left[\frac{\partial L}{\partial z^{[3]}} \right]$$

(1 × 1)

$$\frac{\partial L}{\partial w^{[2]}} = ?$$

$$\frac{\partial z_i^{[2]}}{\partial w_{kj}^{[2]}} =$$

$$\frac{\partial L}{\partial w_{kj}^{[2]}} = \frac{\partial L}{\partial z_i^{[2]}}$$

$$\frac{\partial L}{\partial w_{kj}^{[2]}} =$$

$$\frac{\partial L}{\partial z^{[3]}} \rightarrow (1 \times 10) \quad \begin{matrix} \partial \\ \left[\begin{matrix} w_{11} \\ \vdots \\ w \end{matrix} \right] \end{matrix}$$

$$\frac{\partial L}{\partial b^{[2]}} = \left[\begin{matrix} \frac{\partial L}{\partial z^{[3]}} \cdot (w^{[3]})^T \\ (1 \times 10) \end{matrix} \right] \odot \frac{\partial a^{[2]}}{\partial z^{[2]}}$$

↑
element
wise multiply

$$\frac{\partial L}{\partial w^{[2]}} = ?$$

$$\frac{\partial z_j^{[2]}}{\partial w_{kj}^{[2]}} = a_k^{[1]}$$

$$\frac{\partial L}{\partial w_{kj}^{[2]}} = \frac{\partial L}{\partial z_j^{[2]}} \times \frac{\partial z_j^{[2]}}{\partial w_{kj}^{[2]}} = \frac{\partial L}{\partial z_j^{[2]}} \times \frac{\partial b_j^{[2]}}{\partial w_{kj}^{[2]}}$$

$$\frac{\partial L}{\partial w_{kj}^{[2]}} = \underbrace{\left(\sum_{i=1}^{10} \frac{\partial L}{\partial z_i^{[3]}} w_{ji}^{[3]} \right)}_{\delta_j} \frac{\partial a_j^{[2]}}{\partial z_j^{[2]}} \times a_k^{[1]}$$